Gates in the sky: On ISRO and SpaDeX

Docking technology allows ISRO to think of longer space flights

On December 30, 2024, the Indian Space Research Organisation (ISRO) launched its PSLV-C60 mission. Its primary **payload** was a pair of satellites for the Space Docking Experiment, or SpaDeX, to **demonstrate orbital rendezvous** and **docking**. The **ability** to **execute** it in **orbit** **is** an essential **stepping stone** to more complex missions. The launch was also **hailed** as ISRO ending the year on a high, but SpaDeX reminded us that sophisticated spaceflight missions care little for **arbitrary** deadlines. The satellites successfully **docked** on January 16 after a few **abortive** attempts. They were expected to dock on January 7, which ISRO postponed to January 9, then **brought them close** without docking on January 12 in an **apparent** data-taking effort. It **nixed** the January 9 attempt after the satellites were found to have **drifted** more than expected, prompting measures to **arrest** the **displacement** and **reinitialise** the experiment. Once docked, ISRO began tests to verify if the satellites could exchange electric power, then undock and separate, followed by testing their own payloads that would be spread over two years. The C60 mission also launched the fourth stage of the rocket as an orbital platform. It carried 24 payloads from various ISRO centres and private enterprises testing various technologies. The Vikram Sarabhai Space Centre's Compact Research module for Orbital Plant Studies was able to have cowpea seeds **germinate** in orbit, capturing the popular imagination.

Docking allows **spacefaring** components to be launched separately and **assembled** in space to form a larger **module**. This allows a space agency to plan **interplanetary** missions whose spacecraft are heavier than what the heaviest rockets can launch. Docking is thus a symbolic gateway to new opportunities, with the Chandrayaan-4 **lunar** sample return mission being a good example. In **anticipation**, ISRO loaded the SpaDeX satellites with enough fuel for multiple tries and also continuously collected data. Its own **nervousness** became **evident**, too: after the first two attempts, it **backed down** from its promise to live-stream the successful one. Docking technology has become desirable **thanks to** the **perceived inevitability** of long-duration spaceflight. The **pressure** to lower costs **imposed** by, say, **crewed** missions to Mars or space-mining operations **has** **rendered** ideas such as in-space satellite servicing and orbital resupply platforms, both of which require docking, more **lucrative**. ISRO plans to start launching the 'Bharatiya Antariksh Station' (BAS) later this **decade**. As it **embarks on** a new phase of operations, with V. Narayanan as its new chairman, ISRO should also describe a **coherent** vision for the **ex ante utility** expected of BAS. Without this context, the larger pieces of the Indian space programme and their **purpose** relative to other countries' plans **seem** **adrift**. [Practice Exercise]

- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'.
- Nix (verb) – put an end to; cancel. अंत कर देना

Vocabulary

1. **Docking technology** (noun) – Space-connection system, orbital joining mechanism, spacecraft coupling system, module attachment technology, interlocking devices अंतरिक्ष यान संयोजन तकनीक
2. **Payload** (noun) – Cargo, shipment, freight, load, goods भार/माल
3. **Demonstrate** (verb) – Exhibit, showcase, illustrate, display, prove प्रदर्शन करना
4. **Orbital** (adjective) – Related to orbit, revolving, circling, celestial path-related, space trajectory कक्षा से संबंधित
5. **Rendezvous** (noun) – Meeting, assembly, gathering, convergence, encounter मिलन/बैठक
6. **Docking** (noun) – Coupling, joining, connection, attachment, linking युग्मन
7. **Execute** (verb) – Perform, carry out, implement, accomplish, enact कार्यान्वित करना
8. **Orbit** (noun) – Path, trajectory, course, circuit, revolution कक्षा
9. **Stepping stone** (noun) – Milestone, intermediary stage, advancement, foundation, springboard प्रगति का चरण
10. **Hail** (verb) – Acclaim, celebrate, applaud, acknowledge, commend सराहना करना
11. **Arbitrary** (adjective) – Random, capricious, unplanned, erratic, unpredictable मनमाना
12. **Dock** (verb) – Join, connect, attach, couple, link जोड़ना
13. **Abortive** (adjective) – Failed, unsuccessful, fruitless, ineffective, unproductive विफल
14. **Bring someone close** (phrase) – Move near, approach, draw closer, advance towards, come near पास लाना
15. **Apparent** (adjective) – Evident, obvious, clear, visible, noticeable स्पष्ट
16. **Drift** (verb) – Move aimlessly, float, shift, stray, meander बहकना/बहाव
17. **Arrest** (verb) – Stop, halt, restrain, check, inhibit रोकना
18. **Displacement** (noun) – Movement, shift, relocation, misalignment, repositioning विस्थापन
19. **Reinitialise** (verb) – Restart, reset, reboot, reconfigure, begin again पुनः प्रारंभ करना
20. **Germinate** (verb) – Sprout, grow, bud, develop, begin to grow अंकुरित होना
21. **Spacefaring** (adjective) – Space-exploring, cosmic-traveling, interstellar, celestial-navigating, astronomical exploration अंतरिक्ष यात्रा से संबंधित

22. **Assemble** (verb) – Put together, construct, build, fabricate, compile इकट्ठा करना

23. **Module** (noun) – Unit, component, section, segment, element इकाई

24. **Interplanetary** (adjective) – Between planets, celestial, cosmic, planetary, outer-space अंतर्ग्रहीय

25. **Lunar** (adjective) – Related to the moon, moon-based, selenic, moon-bound, celestial चंद्रमा से संबंधित

26. **Anticipation** (noun) – Expectation, foresight, prediction, eagerness, forethought प्रत्याशा

27. **Nervousness** (noun) – Anxiety, apprehension, unease, tension, worry घबराहट

28. **Evident** (adjective) – Obvious, apparent, clear, noticeable, discernible स्पष्ट

29. **Back down** (phrasal verb) – Withdraw, concede, retract, retreat, yield पीछे हटना

30. **Thanks to** (phrase) – Due to, because of, owing to, as a result of, on account of के कारण

31. **Perceived** (adjective) – Recognized, noticed, regarded, observed, understood समझा हुआ

32. **Inevitability** (noun) – Certainty, unavoidability, necessity, predictability, surety अनिवार्यता

33. **Impose** (verb) – Enforce, apply, mandate, compel, implement थोपना

34. **Crewed** (adjective) – Manned, operated by people, staffed, human-occupied, inhabited मानव-संचालित

35. **Render** (verb) – Make, cause, provide, deliver, perform बनाना/प्रदान करना

36. **Lucrative** (adjective) – Profitable, gainful, rewarding, financially beneficial, advantageous लाभकारी

37. **Decade** (noun) – period of ten years दशक

38. **Embark** (on) (verb) – Start, initiate, begin, launch, set out आरंभ करना

39. **Coherent** (adjective) – Logical, consistent, clear, rational, systematic संगठित/सुसंगत

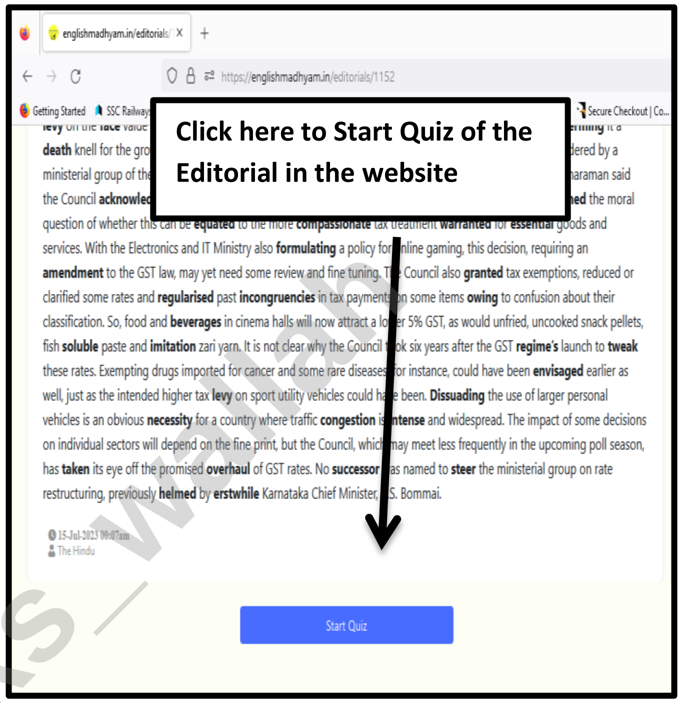
40. **Ex ante utility** (noun) – Expected benefit, pre-planned use, anticipated value, predicted advantage, calculated utility पूर्व लाभ

41. **Adrift** (adjective) – Aimless, directionless, wandering, floating, without purpose उद्देश्यहीन/बहता हुआ

Summary of the Editorial

1. **PSLV-C60 Mission**: ISRO launched PSLV-C60 on December 30, 2024, carrying satellites for the Space Docking Experiment (SpaDeX) to demonstrate orbital rendezvous and docking.
2. **Significance of Docking**: Docking technology is a crucial milestone for ISRO, enabling complex space missions and longer space flights.
3. **Challenges and Success**: After multiple attempts and delays, ISRO successfully docked the satellites on January 16, showcasing resilience and data-driven troubleshooting.
4. **Post-Docking Tests**: ISRO conducted tests on power exchange between docked satellites, undocking, and separation, with additional payload tests planned over two years.
5. **Multi-Payload Mission**: The PSLV-C60 mission also launched the rocket's fourth stage as an orbital platform with 24 payloads from ISRO and private entities, testing various technologies.
6. **Orbital Plant Study**: Cowpea seed germination experiments by the Vikram Sarabhai Space Centre in orbit captured public interest.
7. **Advantages of Docking**: Docking enables the assembly of larger spacecraft in orbit, paving the way for interplanetary missions beyond the limits of current rocket capacities.
8. **Chandrayaan-4 Implications**: Docking technology is crucial for future missions like Chandrayaan-4's lunar sample return, demonstrating its practical application.
9. **Fuel and Data Preparation**: The SpaDeX satellites were equipped with ample fuel for repeated attempts, and continuous data collection aided in refining the docking process.
10. **Cost-Effective Space Operations**: Docking facilitates in-space satellite servicing, orbital resupply platforms, and other technologies, reducing costs for long-duration missions like crewed Mars flights.
11. **Bharatiya Antariksh Station (BAS)**: ISRO plans to launch the BAS by the end of the decade, signaling a shift toward advanced space operations.
12. **ISRO's Nervousness**: The organization avoided live-streaming the final successful docking attempt, indicating caution amidst high stakes.
13. **Leadership Transition**: With V. Narayanan as the new ISRO chairman, the agency enters a phase requiring strategic clarity and ambition.
14. **Global Comparisons**: ISRO needs a coherent vision for the BAS to define its utility and position within global space efforts.
15. **Vision for the Future**: A clear articulation of India's space programme objectives will strengthen its relevance and coordination with international space exploration goals.

Practice Exercise: SSC Pattern Based



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